



Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF  
**COMPUTER SCIENCE AND ENGINEERING**

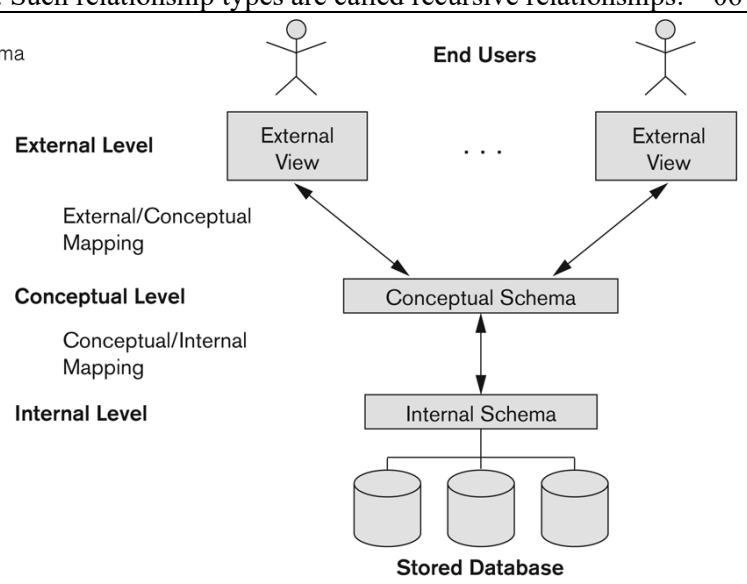
|                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                     |               |           |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|---------------|-----------|
| <b>Date</b>                                                            | 25 <sup>th</sup> November 2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Maximum Marks</b>                | 10+50         |           |
| <b>Course Code</b>                                                     | CD252IA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Duration</b>                     | 20+90 Minutes |           |
| <b>Sem</b>                                                             | V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Faculty: CNS/PD/SB/HR/PT//PHMNV/SNM |               |           |
| <b>Database Management Systems (Common to CS, IS, CD, AI &amp; CY)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                     |               |           |
| <b>Sl.No</b>                                                           | <b>PART-A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>M</b>                            | <b>BT</b>     | <b>CO</b> |
| 1.                                                                     | Identify two tasks performed by "actors on the scene" and "workers behind the scene" in a database environment. Provide examples of job roles for each category.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2                                   | L3            | 1         |
| 2.                                                                     | Which data model balances ease of understanding for end users and its implementation on computer systems? Give an example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2                                   | L3            | 1         |
| 3.                                                                     | Given the scenario of a Library Management system, Transaction record entity that represents the details of books issued to members can be treated as a weak entity. Justify the statement with reason for weakness.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2                                   | L3            | 2         |
| 4.                                                                     | Differentiate between multivalued and composite attributes. Give an example representation for each.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2                                   | L2            | 2         |
| 5.                                                                     | Outline the importance of NULL value. Mention two instances where a value of given attribute can be NULL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2                                   | L2            | 1         |
| <b>Sl. No.</b>                                                         | <b>PART-B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>M</b>                            | <b>BT</b>     | <b>CO</b> |
| 1                                                                      | For a <b>relationship type</b> $R$ among $n$ entity types $E_1, E_2, \dots, E_n$ mathematically define:<br>i) Relationship Set<br>ii) Relationship instances<br>Also determine using an example,<br>iii) Degree of relationship type<br>iv) Relationship as attribute<br>v) Role names in Recursive Relationship.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 10                                  | L2            | 1         |
| 2                                                                      | (a) With the help of a neat diagram, explain the three schema architecture.<br>(b) Using the above architecture, outline the types of data independence.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 06<br>04                            | L2<br>L2      | 1<br>1    |
| 3                                                                      | (a) Explain the characteristics of database approach.<br>(b) Differentiate between the total participation constraint and partial participation constraint. Give example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 06<br>04                            | L2<br>L2      | 1<br>1    |
| 4                                                                      | A Farmer/Consumer Management system needs to be developed. The main objective of this project is to build an application which will help farmers from Indian villages to sell their products to different cities. Farmers wishing to avail the facility can directly register in the system, and sell their product. On the other side, consumers and wholesalers from urban settlements can procure their products from such agricultural colonies. The validity of all such transactions is tabulated for all future references. Requirements could be but not limited to<br>• Upload of the crop to be sold.<br>• Displays a list of the crops and their prices by that farmer.<br>• Ordering any of the produce.<br>• Update logs and transactions on sale.<br>• The quantity and price of all the produce must be known to the farmer at the time of uploading.<br>a) Design an ER diagram for the above requirements. Make suitable assumption wherever required and justify the same explicitly.<br>b) Make suitable assumption to represent all participation types, cardinality ratios. Identify and represent different types of attributes. Justify your answer. | 10                                  | L4            | 2         |
| 5                                                                      | For the above E-R Diagram, Apply the ER to Relational Mapping algorithm to convert ER to Relational Schema. Depict all steps appropriately and justify your answer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 10                                  | L4            | 2         |



Academic year 2024-2025 (Odd Sem)

| DEPARTMENT OF<br>COMPUTER SCIENCE AND ENGINEERING<br>SCHEME AND SOLUTION |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |            |    |    |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------|----|----|
| Date                                                                     | 24th November 2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Maximum Marks | 10+50      |    |    |
| Course Code                                                              | CD252IA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Duration      | 90 Minutes |    |    |
| Sem                                                                      | V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Faculty:      |            |    |    |
| CIE-I                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |            |    |    |
| Database Management Systems (Common to CS, IS, CD, AI & CY)              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |            |    |    |
| Sl.No                                                                    | PART-A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               | M          | BT | CO |
| 1.                                                                       | <b>Tasks of Actors on the Scene:</b><br>1. Querying the database for specific information (e.g., retrieving customer data).<br>2. Updating or modifying records in the database (e.g., editing employee information).<br>Examples: Data Analysts, Bank Clerks, Sales Executives.<br><b>Tasks of Workers Behind the Scene:</b><br>1. Ensuring the database system is running efficiently (e.g., tuning database performance).<br>2. Implementing security measures to protect the database (e.g., setting user access permissions).<br>Examples: Database Administrators (DBAs), System Engineers, Security Specialists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               | 2          | L3 | 1  |
| 2.                                                                       | <b>Representational (or Implementation) Data Model.</b><br>Example: Model which organizes data into tables (relations) with rows and columns, allowing users to query and manipulate data using SQL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               | 2          | L3 | 1  |
| 3.                                                                       | It depends on the combination of the <b>Member ID</b> (from the <b>Member</b> entity) and the <b>Book ID</b> (from the <b>Book</b> entity) for its identification.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               | 2          | L3 | 2  |
| 4.                                                                       | Composite attributes can be divided into smaller subparts, which represent more basic attributes with independent meanings.<br>Multivalued attributes are those for which different users may have different values for the same attribute. It may have lower and upper bounds to constrain the number of values allowed for each individual entity. – 1 mark<br>Representation for both: 1 mark                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               | 2          | L2 | 2  |
| 5.                                                                       | In some cases, a particular entity may not have an applicable value for an attribute. For such situations, a special value called NULL is created. An address of a single-family home would have NULL for its Apartment_number attribute, and a person with no college degree would have NULL for College_degrees. NULL can also be used if we do not know the value of an attribute for a particular entity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               | 2          | L2 | 1  |
| Sl. No.                                                                  | PART-B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               | M          | BT | CO |
| 1                                                                        | A relationship type R among n entity types E1, E2, ..., En defines a set of associations—or a relationship set—among entities from these entity types.<br>Mathematically, the relationship set R is a set of relationship instances ri, where each ri associates n individual entities (e1, e2, ..., en), and each entity ej in ri is a member of entity set Ej, 1 ≤ j ≤ n. Hence, a relationship set is a mathematical relation on E1, E2, ..., En; alternatively, it can be defined as a subset of the Cartesian product of the entity sets E1 × E2 × ... × En. Each of the entity types E1, E 2, ..., En is said to participate in the relationship type R; similarly, each of the individual entities e1, e2, ..., en is said to participate in the relationship instance ri = (e1, e2, ..., en). – 040 marks<br><br>iii) An example of a ternary relationship is SUPPLY, where each relationship instance ri associates three entities—a supplier s, a part p, and a project j—whenever s supplies part p to project j. Relationships can generally be of any degree, but the ones most common are binary relationships.<br>iv) Relationships as Attributes: Consider the WORKS_FOR relationship type. One can think of an attribute called Department of the EMPLOYEE entity type, where the value of Department for each EMPLOYEE entity is (a reference to) the DEPARTMENT entity for which that employee works. Hence, the value set for this Department attribute is the set of all DEPARTMENT entities, which is the DEPARTMENT entity set. |               | 10         | L2 | 1  |

Academic year 2024-2025 (Odd Sem)

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|---|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|--------|
|   |     | v) Role names are not technically necessary in relationship types where all the participating entity types are distinct, since each participating entity type name can be used as the role name. However, in some cases the same entity type participates more than once in a relationship type in different roles. In such cases the role name becomes essential for distinguishing the meaning of the role that each participating entity plays. Such relationship types are called recursive relationships. – 06 marks                                                                                                                                                                                                                                                                                                                                                                                                                              |          |          |        |
| 2 | (a) | <p><b>Figure 2.2</b><br/>The three-schema architecture.</p>  <p>Fig -2 marks, Expln: 4 marks</p> <p><b>Logical data independence</b> is the capacity to change the conceptual schema without having to change external schemas or application programs. We may change the conceptual schema to expand the database (by adding a record type or data item), to change constraints, or to reduce the database (by removing a record type or data item). <b>Physical data independence</b> is the capacity to change the internal schema without having to change the conceptual schema. Hence, the external schemas need not be changed as well. Changes to the internal schema may be needed because some physical files were reorganized—for example, by creating additional access structures—to improve the performance of retrieval or update.</p> <p>2X2 =4</p> | 06<br>04 | L2<br>L2 | 1<br>1 |
| 3 | (a) | <p>List -2M<br/>Explanation 4M</p> <ul style="list-style-type: none"> <li>• Self-describing nature of a database system</li> <li>• ■ Insulation between programs and data, and data abstraction</li> <li>• ■ Support of multiple views of the data</li> <li>• ■ Sharing of data and multiuser transaction processing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 06       | L2       | 1      |
|   | (b) | <p>Participation Constraints Definition 2M<br/>Example 2M<br/>Total Participation<br/>Partial Participation</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 04       | L2       | 1      |
| 4 | (a) | <p><b>Entities and Attributes</b></p> <p>1. Farmer</p> <ul style="list-style-type: none"> <li>○ <b>Attributes:</b> <ul style="list-style-type: none"> <li>▪ FarmerID (Primary Key)</li> <li>▪ Name</li> <li>▪ ContactNumber</li> <li>▪ Address</li> <li>▪ BankDetails</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 10       | L4       | 2      |

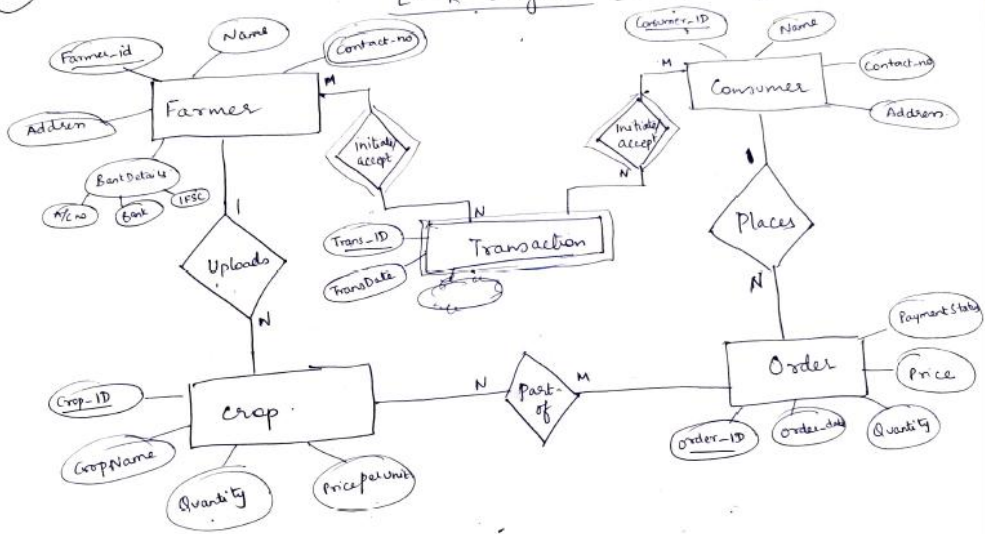


Academic year 2024-2025 (Odd Sem)

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|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
|  |  | <ul style="list-style-type: none"><li>2. <b>Consumer/Wholesaler</b><ul style="list-style-type: none"><li>○ <b>Attributes:</b><ul style="list-style-type: none"><li>▪ ConsumerID (Primary Key)</li><li>▪ Name</li><li>▪ ContactNumber</li><li>▪ Address</li></ul></li></ul></li><li>3. <b>Crop/Produce</b><ul style="list-style-type: none"><li>○ <b>Attributes:</b><ul style="list-style-type: none"><li>▪ CropID (Primary Key)</li><li>▪ CropName</li><li>▪ QuantityAvailable</li><li>▪ PricePerUnit</li></ul></li></ul></li><li>4. <b>Order</b><ul style="list-style-type: none"><li>○ <b>Attributes:</b><ul style="list-style-type: none"><li>▪ OrderID (Primary Key)</li><li>▪ OrderDate</li><li>▪ QuantityOrdered</li><li>▪ TotalPrice</li><li>▪ PaymentStatus</li></ul></li></ul></li><li>5. <b>Transaction</b><ul style="list-style-type: none"><li>○ <b>Attributes:</b><ul style="list-style-type: none"><li>▪ TransactionID (Primary Key)</li><li>▪ TransactionDate</li><li>▪ FarmerID (Foreign Key)</li><li>▪ ConsumerID (Foreign Key)</li><li>▪ OrderID (Foreign Key)</li></ul></li></ul></li></ul>                                                                                                                  |  |  |  |
|  |  | <p><b>Relationships</b></p> <ul style="list-style-type: none"><li>1. <b>Farmer uploads Crop/Produce</b><ul style="list-style-type: none"><li>○ <b>Relationship:</b> One Farmer can upload multiple Crops, but each Crop belongs to one Farmer.</li><li>○ <b>Type:</b> One-to-Many</li></ul></li><li>2. <b>Consumer places Order</b><ul style="list-style-type: none"><li>○ <b>Relationship:</b> One Consumer can place multiple Orders, but each Order is placed by one Consumer.</li><li>○ <b>Type:</b> One-to-Many</li></ul></li><li>3. <b>Crop is part of Order</b><ul style="list-style-type: none"><li>○ <b>Relationship:</b> One Order can contain multiple Crops, and each Crop can be part of multiple Orders.</li><li>○ <b>Type:</b> Many-to-Many (resolved through an associative entity, e.g., <b>OrderDetails</b>)</li></ul></li><li>4. <b>Farmer and Consumer/Wholesaler involved in Transaction</b><ul style="list-style-type: none"><li>○ <b>Relationship:</b> Each Transaction involves one Farmer and one Consumer.</li><li>○ <b>Type:</b> Many-to-Many (can be resolved using the <b>Transaction</b> entity).</li></ul></li></ul> <p><b>ER Diagram – 07 marks</b><br/><b>Cardinality ratio – 03 marks</b></p> |  |  |  |



Academic year 2024-2025 (Odd Sem)

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|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|----|
|   | <p>Q4</p> <p>E-R Diagram (Sample Only)</p>  <p>Note: Transaction can also be a relation with attributes later converted into a relation/Entity</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |    |
| 5 | <p>(a) <b>Relational Tables – 07 marks</b></p> <ol style="list-style-type: none"> <li><b>Farmer</b> <ul style="list-style-type: none"> <li>FarmerID (Primary Key)</li> <li>Name</li> <li>ContactNumber</li> <li>Address</li> <li>BankDetails</li> </ul> </li> <li><b>Consumer</b> <ul style="list-style-type: none"> <li>ConsumerID (Primary Key)</li> <li>Name</li> <li>ContactNumber</li> <li>Address</li> </ul> </li> <li><b>Crop</b> <ul style="list-style-type: none"> <li>CropID (Primary Key)</li> <li>FarmerID (Foreign Key, references Farmer.FarmerID)</li> <li>CropName</li> <li>QuantityAvailable</li> <li>PricePerUnit</li> </ul> </li> <li><b>Order</b> <ul style="list-style-type: none"> <li>OrderID (Primary Key)</li> <li>ConsumerID (Foreign Key, references Consumer.ConsumerID)</li> <li>OrderDate</li> <li>PaymentStatus</li> </ul> </li> <li><b>OrderDetails</b> (Associative table for Many-to-Many relationship between Order and Crop) <ul style="list-style-type: none"> <li>OrderID (Foreign Key, references Order.OrderID)</li> <li>CropID (Foreign Key, references Crop.CropID)</li> <li>QuantityOrdered</li> <li>TotalPrice</li> </ul> </li> <li><b>Transaction</b> <ul style="list-style-type: none"> <li>TransactionID (Primary Key)</li> <li>FarmerID (Foreign Key, references Farmer.FarmerID)</li> <li>ConsumerID (Foreign Key, references Consumer.ConsumerID)</li> <li>OrderID (Foreign Key, references Order.OrderID)</li> </ul> </li> </ol> | 10 | 4 | L2 |

Academic year 2024-2025 (Odd Sem)

○ TransactionDate

**Relational Schema Diagram Representation – 03 marks**

- **Primary Keys (PK):** Underlined in each table.
- **Foreign Keys (FK):** Marked as references to their respective tables.
- Relationships are shown using lines/arrows connecting FK columns to their referenced PK columns.

Representation of the above diagram step by step as per algorithm

